DATA301 Individual Report

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Executive Summary

Background

Data Description

The dataset is the Steam Games dataset published on Kaggle by Artemiy Ermilov. It is the cleaned March 2025 version, with 89,618 games and 47 variables, covering releases from April 1997 to March 2025. Each row is one game.

My question uses five variables:

- genre - categorical, multi-label per game (a game can carry several genre tags)
- price - continuous, float64, base game price
- required_age - minimum age rating
- pct_pos_total - percentage of total positive reviews. This is my response variable.
- num_reviews_total - total review count. Used only for filtering, not as a predictor

Two data quality issues shape how these variables can be used. First, Steam sets both pct_pos_total and num_reviews_total to a -1 sentinel for games with fewer than 10 reviews. This is not missing data in the usual sense, so it needs filtering rather than imputation. Following the group EDA, I apply a 30-review minimum, which leaves 34,583 of the 89,618 games (38.6%). Second, 212 games with no genre listed were dropped, since genre is the factor of interest and cannot be imputed.

On top of the review filter, the group excluded six non-genre content descriptors (Violent, Gore, Nudity, Sexual Content, Free to Play, Early Access), taking 27 raw genre tags down to 21 genres. A minimum of 100 games per genre was then applied, leaving 10 established genres with stable estimates. price and required_age have no missing values.

One structural point still needs resolving with my supervisor. The genres column holds multiple tags per game. The group EDA exploded this into one row per game-genre pair for descriptive summaries. My regression needs one row per game, so I will likely switch to a single primary_genre column. I'm confirming this with Frery before finalizing.

Ethics, Privacy and Security

The data was collected by scraping the Steam store page, the Steam API, and public user metrics such as reviews and scores. All of this is publicly available information, so there are few concerns around the collection process itself.

The dataset contains copyrighted material, including game titles, descriptions, and reviews. My analysis only uses structured numeric and categorical fields (genre, price, required_age, pct_pos_total), so I do not reproduce or republish any review text or game descriptions.

There is inherent bias in review-based metrics. Most people who own a game never leave a rating, and the few who do may not represent the wider player base. This is directly relevant to my response variable, pct_pos_total, since it is built entirely from that self-selected group. Popularity also confounds this: bigger games accumulate more reviews and more stable percentages, while niche genres are more volatile. The 100-games-per-genre threshold reduces this at the genre level, but review-count imbalance within a genre remains a limitation worth naming in my discussion.

No Steam personal information appears in the variables I use, and since I do not process review text, I have no exposure to personal details users might include in free-text reviews. Project materials stay within the same access controls as the group: private GitHub repo, access-restricted Google Docs, and group-only Trello and Messenger channels.

Exploratory Data Analysis

Detailed Analysis Result

Conclusion and Recommendations

References

Appendices